

SCHOOL OF ENGINEERING

Diploma in Electronics & Computer Engineering (EGDF20)

Wireless Communication & Networking (EGE321)

Tutorial 1: RF Fundamentals

- 1) *Define wireless communication. What are the advantages and disadvantages of wireless communication?*

Wireless communication involves the transmission of information over a distance without the need of wires, cables or any other forms of electrical conductors in between the transmitter and receiver.

Advantages

- A wireless communication network is a solution in areas where cables are impossible to install.
- Wireless networks are cheaper to install and maintain.
- Flexible

Disadvantages

- Has security vulnerabilities.
- High cost for setting the infrastructure.
- Unlike wired communication, wireless communication is influenced by physical obstructions, climatic conditions and interference from other wireless devices.

- 2) *Name and explain the three types of transmission modes in data communication and their applications.*

YouTube link

[Simplex, Half duplex and Full duplex, the different types of transmission modes - YouTube](#)

Simplex

Information travels in **one direction** only

TV and Radio broadcast

Half duplex

Information travels in **both directions** but one direction at a time => at any one time **only one party can speak and the other party must list**

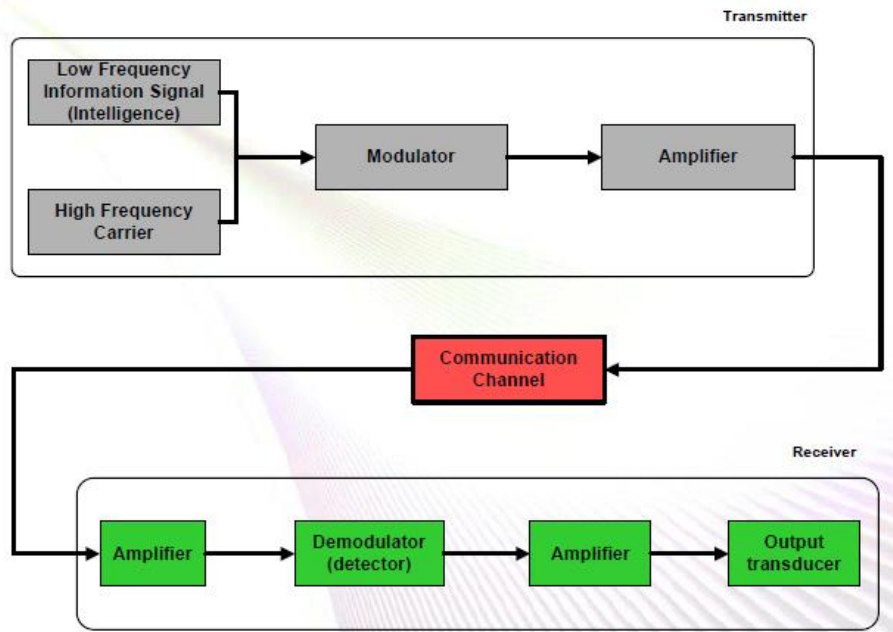
Walkie Talkie

Full duplex

Information travels in **both directions simultaneously** => both parties can speak at the same time

Mobile Phone

- 3) Draw the basic building block of an RF system and explain briefly the function of each block.



Low frequency information signal – The signal that you intend to send over

High frequency carrier – The purpose of the carrier is usually either to transmit the information through air or to allow several carriers at different frequencies to share a common physical transmission medium.

Modulator – An RF modulator (or radio frequency modulator) is an electronic device whose input is a baseband signal which is used to modulate a radio frequency source.

Amplifier - An amplifier functions by using electric power from a power supply to increase the amplitude of the voltage or current signal. An amplifier is effectively the opposite of an attenuator: while an amplifier provides gain, an attenuator provides loss.

Communication channel – A medium through which a message is transmitted to its intended audience.

Demodulator – A device used to recover the original modulating signal from a modulated wave. A demodulator is also known as a detector.

Output transducer - A device which can convert information from one physical form to another. The output transducer transfers signals out of the electrical domain and into the domain that can be perceived by one of the five human senses.

- 4) How fast does a radio wave travel through free space? What is the approximate velocity of a radio wave as it travels through free space?

Speed of light. 3×10^8 m/s

- 5) How does the wavelength of a radio wave relate to its frequency? What is the frequency of a signal when its wavelength λ is 0.6 m and the velocity of propagation is c ? If the wavelength is 0.4 cm, what is the frequency?

The wavelength gets shorter as the frequency increases. In another word, frequency and wavelength are inversely proportional to each other.

$$c = f \lambda$$

$$f = c/\lambda$$

$$f = 3 \times 10^8 / 0.6$$
$$= 500 \text{ MHz}$$

if the is $\lambda = 0.4 \text{ cm} = 0.004 \text{ m}$, (Convert to m)

$$f = 3 \times 10^8 / 0.004 = 75 \text{ GHz}$$

- 6) A UWB signal occupies the frequency range from 3.1 to 10.6 MHz. What is its bandwidth?

$$\text{BW} = \text{highest frequency} - \text{lowest frequency}$$
$$= 10.6 - 3.1 \text{ MHz}$$
$$= 7.5 \text{ MHz}$$